

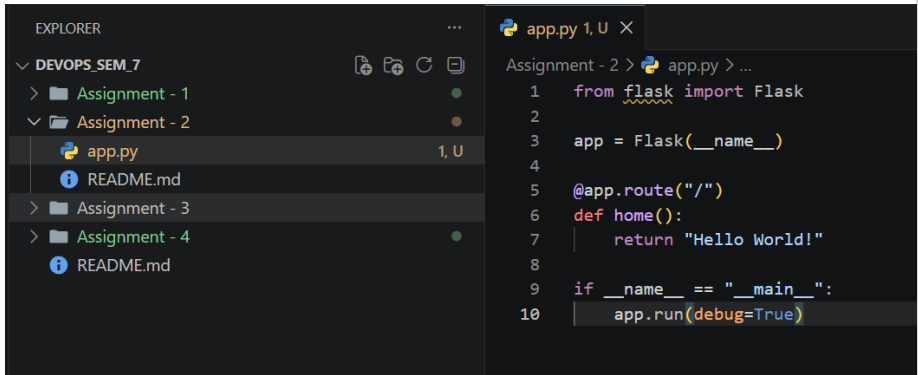


# SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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## Department of Computer Science and Engineering DevOps Lab - L3

| Lab Assignment No: 2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Name of Student          | Aryan Sheladia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PRN                      | 23070122202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Batch                    | 2023 - 2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Class                    | CSE – B1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Academic Year & Semester | 2026 - 2027, 7th Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Title of Assignment      | Git Branching & Merging                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Implementation & Theory  | <div><pre>1 from flask import Flask 2 3 app = Flask(__name__) 4 5 @app.route("/") 6 def home(): 7     return "Hello World!" 8 9 if __name__ == "__main__": 10     app.run(debug=True)</pre></div> <p>Created the Assignment - 2 directory and developed a basic Flask "Hello World" application. This serves as the initial codebase for demonstrating Git branching and collaboration.</p> <div><pre>PS E:\DevOps_SEM_7&gt; git branch * main PS E:\DevOps_SEM_7&gt;</pre></div> <p>Verified the current Git branch before creating a new feature branch. This ensures that the new feature is developed from the latest state of the <b>main</b> branch, following the standard Git workflow.</p> |

```

PS E:\DevOps_SEM_7> git checkout -b feature/user-auth
Switched to a new branch 'feature/user-auth'
PS E:\DevOps_SEM_7> git branch
* feature/user-auth
  main
PS E:\DevOps_SEM_7>

```

Created and switched to a new feature branch named **feature/user-auth** using Git. Feature branches isolate new development from the main branch, allowing changes to be implemented, tested, and reviewed independently before merging.

```

app.py 1, U X
Assignment - 2 > app.py > ...
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      print("User Authentication Feature Added")
8      return "Hello World!"
9
10 if __name__ == "__main__":
11     app.run(debug=True)

```

Modified the Flask application by adding a print statement to simulate the implementation of a new feature. This demonstrates how changes can be developed independently on a feature branch before being integrated into the main branch.

```

PS E:\DevOps_SEM_7> git status
On branch feature/user-auth
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Assignment - 2/app.py

nothing added to commit but untracked files present (use "git add" to track)

```

Executed the git status command to verify that Git detected the modification made to the Flask application. The output confirms that the updated file is ready to be staged for version control.

```

nothing added to commit but untracked files present (use "git add" to track)
● PS E:\DevOps_SEM_7> git add .
● PS E:\DevOps_SEM_7> git status
On branch feature/user-auth
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Assignment - 2/app.py

○ PS E:\DevOps_SEM_7>

```

Staged the modified application using `git add .`, preparing the changes for the next commit. Staging allows developers to review and selectively include changes in a commit.

```

● PS E:\DevOps_SEM_7> git commit -m "Added user authentication feature"
[feature/user-auth 18c45a5] Added user authentication feature
 1 file changed, 11 insertions(+)
   create mode 100644 Assignment - 2/app.py
○ PS E:\DevOps_SEM_7>

```

Created a commit with a meaningful message describing the implemented feature. Each commit acts as a permanent snapshot of the project, making it easier to track changes and maintain version history.

```

● PS E:\DevOps_SEM_7> git push -u origin feature/user-auth
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 558 bytes | 279.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:   https://github.com/AryanSheladia/DevOps_SEM_7/pull/new/feature/user-auth
remote:
To https://github.com/AryanSheladia/DevOps_SEM_7
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.

```

Pushed the `feature/user-auth` branch to the remote GitHub repository. This makes the feature available for collaboration, review, and eventual merging into the main branch.



feature/user-auth

DevOps\_SEM\_7 / Assignment - 2



AryanSheladia Added user authentication feature

This branch is 1 commit ahead of and 1 commit behind main .

Name



..



README.md



app.py

README.md